

Data Lake

Repository for large volumes of structured and unstructured data used for analytics and processing.

Concept

Data Lakes store varied data formats, often before final analytical modeling.

Risks

Schema drift, duplicates, incorrect partitions, permissions, late data and ingestion quality are common risks.

QA

Validate ingestion, metadata, partitions, volume, transformation rules and access.

Lake vs Warehouse

Data Lakes favor flexibility/raw data; Data Warehouses usually organize modeled data for analytics consumption.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

QA tip

Test metadata and partitions too; existing files do not guarantee a usable dataset.

Official resources

Site oficial: <https://aws.amazon.com/what-is/data-lake/>